

Social Mode. Full Spec.

The complete, code-accurate reference for the BAC estimator, impairment escalation, transportation prompts, recovery flow, and UI surfaces that power AForce OS Social Mode.

Version 1.0 · Generated from production source
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Overview

Social Mode is the subscriber-only "drink smart on nights out" experience inside AForce OS. It tracks per-drink hydration impact in real time, estimates BAC, escalates transportation safety prompts, and runs an 8-hour morning recovery protocol. The voice is **calm, protective, never preachy** — it never says "safe to drive," never moralizes, and never asserts legal compliance.

1 ACCESS & PAYWALL

TIER	BEHAVIOR
<code>core</code> (free)	Paywall card "Drink smart on nights out" with feature list + UPGRADE TO UNLOCK CTA pointing to Store. Brand purple <code>#7C5CFF</code> .
Athlete+ (any paid tier)	Full <code>SocialModeSheet</code> rendered inline on the Social tab, permanently mounted. Closing returns to Home.

Source of truth: Stripe, mirrored via webhook. Any `planId` other than `core` unlocks Social Mode.

2 DRINK CATALOG

TYPE	DEFAULT OZ	DEFAULT ABV	DECAY ×	ACTIVE WINDOW	SUGAR LOAD	RISK WEIGHT
beer	12	5.0%	1.15×	30 min	3	1.0
wine	5	12.5%	1.20×	30 min	4	1.3
cocktail	8	14.0%	1.30×	30 min	8	1.6
liquor	1.5	40.0%	1.35×	30 min	1	1.8
hard_seltzer	12	5.0%	1.15×	30 min	1	1.0
custom	6	12.0%	1.25×	30 min	4	1.4

Decay multipliers are taken from the **high end** of each range — the user is opting *in* to Social Mode and wants the system to err on the protective side, not flatter them.

3 BAC ESTIMATION — WIDMARK APPROXIMATION

```
gramsAlcohol = Σ ( drink.oz × ML_PER_OZ × (drink.abv/100) × 0.789 )
bodyMassG    = bodyWeightLbs × 453.592    (floored to 80 lbs)
r             = 0.68 (male / unspecified) | 0.55 (female)
foodFactor    = 0.92 if ateRecently else 1.0
bacRaw        = (gramsAlcohol / (bodyMassG × r)) × 100 × foodFactor
elapsedH      = hours since first drink
bacCurrent    = max(0, bacRaw − 0.015 × elapsedH)
```

Output: a range, never a single number.

FIELD	DERIVATION
<code>rangeLow</code> / <code>rangeHigh</code>	point estimate ± 0.01 (Widmark's ±15–20% literature error)
<code>trend</code>	current – BAC 15 min ago; $ \Delta < 0.005$ = <code>steady</code> , else <code>rising</code> / <code>falling</code>
<code>confidence</code>	<code>high</code> if sex provided AND ≥50% drinks have explicit oz/abv AND ≤8 drinks; else <code>medium</code> ; <code>low</code> once >8 drinks
<code>timeToClearMinutes</code>	$(bac - 0.005) / 0.015 \times 60$, rounded up to nearest 5 min (conservative)
<code>notes</code>	i18n keys: <code>default_sex</code> , <code>default_strength</code> , <code>food_factored</code> , <code>many_drinks</code>

Constants: `ETHANOL_DENSITY` = 0.789 g/mL · `ML_PER_OZ` = 29.5735 · `LBS_TO_G` = 453.592 · `ELIMINATION` = 0.015 /h · `CLEAR_THRESHOLD` = 0.005 .

4 IMPAIRMENT ESCALATION MATRIX

Mapping uses the **midpoint of the BAC range** so the trend is smooth across refreshes.

BAC MIDPOINT	LEVEL	SAFETY CARD	STOP DRINKING?	COACH LINE
< 0.030	LOW	hidden	no	standard hydration / pacing copy
0.030 - 0.049	ELEVATED	hidden	no	standard hydration / pacing copy
0.050 - 0.079	MODERATE	shown — caution	no	"Plan a ride before your next drink"
0.080 - 0.119	HIGH	shown — warning	yes	"Do not drive. Use a rideshare."
≥ 0.120	CRITICAL	shown — critical	yes	"Stop alcohol intake. Recovery req."

The safety card is hidden below MODERATE so users only see the legal-protection language when it actually applies. The "stop drinking" sub-prompt only appears at HIGH/CRITICAL.

5 RECOVERY MODE

- Triggered when the user taps **End Night** in the Social sheet — engine writes `socialMode.endedAt` .
- `RECOVERY_WINDOW_MS = 8 hours` . Rollup flips `inRecoveryWindow = true` for that window.
- During recovery, `RecoveryModeCard` renders:
 - Estimated time until BAC clears (`bac.timeToClearMinutes`)
 - 3-step morning protocol:** water → AForce RTD → 7+ hours sleep
 - Standard disclaimer pair
- Home banner shifts to **amber** and reads `social.recovery_active` .
- Rollup returns `null` when Social Mode is neither active nor in recovery window.

6 ORB OVERLAY

`StatusPulseOrb` accepts an optional `socialOverlay?: { alcoholLoad: number; unstable: boolean }` :

- `alcoholLoad ∈ [0, 1]` — derived from the active decay multiplier; pushes a subtle **violet** outer ring.
- `unstable = true` when impairment is HIGH/CRITICAL — ring flips **crimson** and pulses faster.
- Overlay is purely additive; never replaces the base hydration gradient.

7 VOICE & COPY RULES

NEVER: "stop drinking," "you should not," "irresponsible," "safe to drive," promise a specific BAC at a specific future time, assert legal compliance, give medical advice.

STRONGEST positive language allowed: "your estimate is currently low."

- Escalation verbs** (protective, describe outcomes, not character): *plan a ride, switch to water, arrange safe transport, stop alcohol intake*
- Headlines short. Bodies 1-2 sentences max.

8 MANDATORY DISCLAIMER

Every surface that renders a BAC value, impairment level, transport prompt, or recovery time **must** render the standard pair:

Estimate only · Not a legal or medical determination.

i18n keys: `social.estimate_only` , `social.not_legal_medical` . Translated in **en, es, fr, de, pt, it** .

9 DATA MODEL

```
type DrinkType = 'beer'|'wine'|'cocktail'|'liquor'|'hard_seltzer'|'custom'
```

```
DrinkLog {  
  id, type, loggedAt: Date,  
  multiplier,           // hydration decay multiplier  
  hydrated: boolean | null, // null = post-drink prompt pending  
  abv?, oz?             // fall back to catalog defaults  
}
```

```
SocialModeState {  
  active, startedAt, drinks[],  
  lastHydrationPromptAt?, endedAt?,  
  sex?: 'male'|'female'|'unspecified',  
  ateRecently?: boolean  
}
```

```
BACEstimate      { rangeLow, rangeHigh, trend, confidence,  
                  timeToClearMinutes, notes[] }  
ImpairmentRiskState { level, bacMidpoint }  
TransportationSafetyPrompt {  
  show, severity, titleKey, bodyKey, disclaimerKey, stopDrinking  
}  
RecoveryRecommendation { stepKeys[], timeToClearMinutes,  
                          morningEstimateKey }  
HangoverRisk      { level: 'LOW'|'MODERATE'|'HIGH'|'CRITICAL',  
                  score: 0-100, reasons[] }
```

10 ENGINE ROLLUP OUTPUT (PER TICK)

buildSocialRollup(state, now) returns either null or:

```
{  
  active,  
  inRecoveryWindow,  
  drinkCount,  
  hangoverRisk,  
  alcoholMultiplier, // active decay multiplier or 1 when inactive  
  bac,  
  impairment,  
  transportation  
}
```

This is what the UI (SocialModeSheet , SocialModeBanner , SocialSafetyCard , home orb) all read from.

11 UI SURFACE INVENTORY

COMPONENT	ROLE
app/(tabs)/social.tsx	Tab entry — paywall for free, full sheet for paid
components/SocialModeSheet.tsx	Full session UI: activate, log drinks, BAC strip, hydration prompts, end night
components/SocialModeBanner.tsx	Home-screen amber/violet banner indicating active session or recovery
components/SocialSafetyCard.tsx	Transport safety card (only at MODERATE+)
components/StatusPulseOrb.tsx	Hydration orb with optional violet/crimson social ring
screens/JournalScreen.tsx	Past-night summaries

12 PINNED TEST INVARIANTS

These live in services/__tests__/bacEstimation.test.ts and must continue to hold after any engine change:

- Zero drinks → zero BAC, zero clear time.
- One beer → LOW band.
- Four quick liquor shots → at least MODERATE.
- Trend rising right after drinks; falling after long elimination.
- timeToClearMinutes non-negative and a multiple of 5.
- Confidence degrades when sex is unspecified.
- Food intake softens the estimate vs empty stomach.
- Safety prompt hidden at LOW / ELEVATED.
- Safety prompt escalates to "do not drive" at HIGH and CRITICAL.
- Disclaimer key always returned.

